

CONVEYOR BELT WITH SENSORS SYSTEM

Description: A Conveyor Belt with Sensors is an automated material handling system used in industries to transport, detect, count, and sort products efficiently. Sensors monitor the movement of objects and provide feedback to the controller for automated operation.

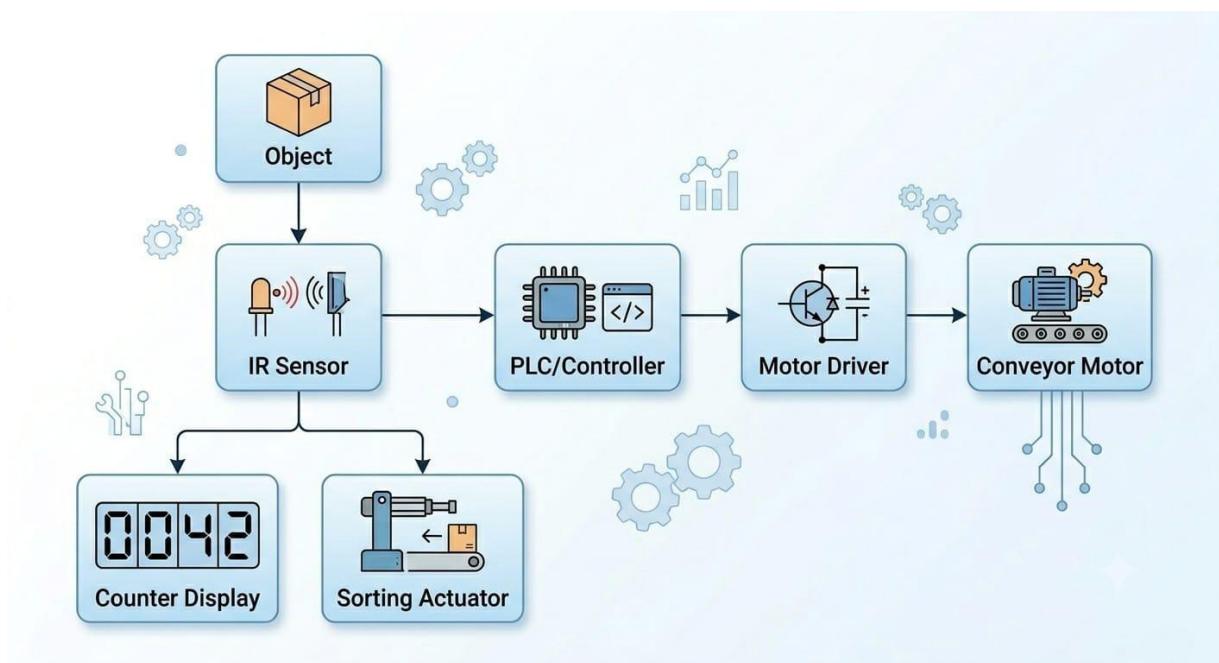
Working Principle:

- ✓ Objects are placed on the conveyor belt.
- ✓ An IR sensor detects the presence of an object.
- ✓ The controller processes the sensor signal.
- ✓ The conveyor motor moves the object to the next stage.
- ✓ The system can count, sort, or reject defective products automatically.

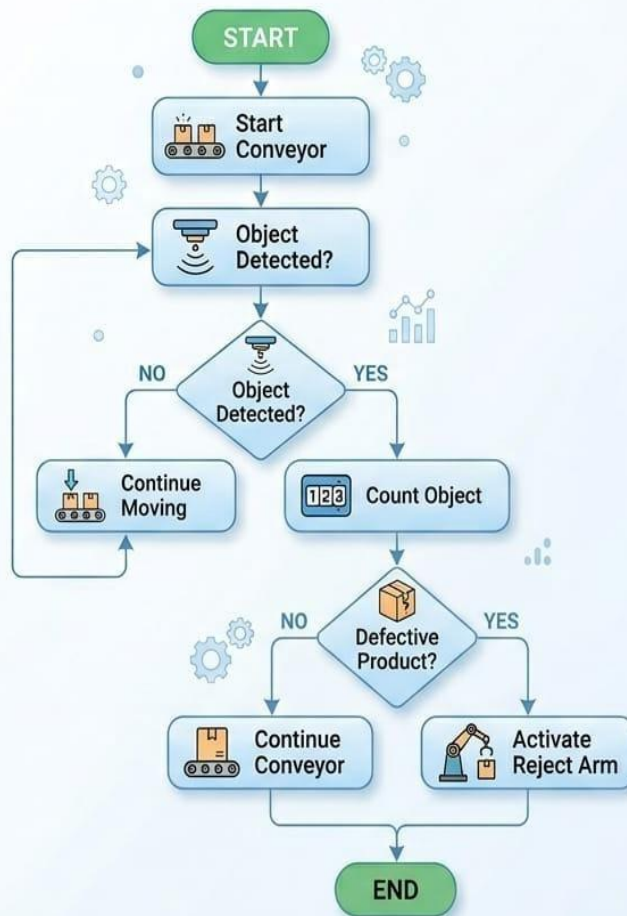
Components Used:

- ✓ IR Sensor
- ✓ PLC/Microcontroller
- ✓ Conveyor Belt Motor
- ✓ Motor Driver
- ✓ Counter Display
- ✓ Sorting Actuator
- ✓ Power Supply

Block Design:



Flowchart:



Advantages:

- ✓ Reduces manual labour
- ✓ Increases production efficiency
- ✓ Improves accuracy in counting and sorting
- ✓ Suitable for industrial automation
- ✓ Reduces operational time